

Age at first marriage in Malawi: a Bayesian multilevel analysis using a discrete time-to-event model

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Summary. The paper presents a hierarchical discrete time survival model for the analysis of the 2000 Malawi Demographic and Health Survey data to assess the determinants of transition to marriage among women in Malawi. The model explicitly accounts for the unobserved heterogeneity by using family and community random effects with cross-level correlation structure. A nonparametric technique is used to model the base-line discrete hazard dynamically. Parameters of the model are computed by using a Markov chain Monte Carlo algorithm. The results show that rising age at marriage is a combination of birth cohort and education effects, depends considerably on the family and to some extent on the community in which a woman resides and the correlation between family and community random effects is negative. These results confirm a downward trend in teenage marriage and that raising women's education levels in sub-Saharan Africa has the beneficial effect of increasing age at marriage, and by implication reducing total fertility rates. The negative correlation between family and community random effects has policy implications in that targeting communities with an intervention to increase age at first marriage may not necessarily yield reduced fertility levels in individual families. A campaign that is geared towards individual families would achieve the desired goals. Overall, the findings point to the need for the Government in Malawi to enact public policies which are geared at vastly improving women's education at higher levels. The variation in marriage rates over families poses problems in delivering the policy, since particular policies must be devised for specific groups of families to accomplish the required social and health objectives.

Keywords: Adolescent marriage and fertility; Gibbs sampling; Sub-Saharan Africa

1. Introduction

Although a decline in fertility has now been well established in sub-Saharan Africa, the fertility still remains high in many countries of the region. The most common measure of fertility is the total fertility rate, which is defined as the average number of births that a woman would have if she survived to the end of her childbearing years. Currently, the total fertility rate for Malawi is 6.3 and the values for the census years 1977, 1987 and 1998 were 7.6, 7.4 and 6.2 respectively (National Statistical Office: Malawi and Macro International, 2000), showing a modest decline. These figures are among the highest in sub-Saharan Africa and in the world, where the standards would generally be under 4 (Siu-Man and Boachang, 1995; Caldwell and Caldwell, 2003). The level of contraceptive use, which is generally low among the youngest and oldest childbearing women, is a major factor affecting levels of fertility in sub-Saharan Africa

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(Kirk and Pillet, 1998; Zlidar and Gardner, 2001). None-the-less, other factors such as levels of education for women, religion and timing of childbearing and marriage are also important determinants of changes in fertility in the region (Harwood-Lejeune, 2000; Gupta and Mahy, 2003). The Government of Malawi's policy makers have been striving to bring the total fertility rate to within acceptable levels. Improvements in family planning and health care have been made while school intakes and employment opportunities for women have also improved.

In this paper, we investigate the determinants of the timing of marriage among women in Malawi. This study is important as the timing of marriage potentially affects the reproductive lifespan and hence has an influence on fertility. By studying variations in age at first marriage, fertility levels across populations and trends within populations can be explained (Mareno, 1991). The pathways through which age at first marriage affects fertility, though well known, are complex. Delayed marriage reduces the proportion of ever-married women who have relatively higher fertility compared with the unmarried (Letamo, 1996). In Malawi, the proportion of the ever-married women in the age group 15–19 years reduced from 48% to 38% in the periods from 1977–1980 to 1989–1992 (National Statistical Office: Malawi and Macro International, 2000), a downward trend which has been confirmed earlier in a comparative study of sub-Saharan African and other developing countries (US Bureau of the Census, 1996). Recently, Harwood-Lejeune (2000) has also shown that rising age at marriage contributes between a sixth and a third of the decline in fertility in nine sub-Saharan countries, including Malawi. But these effects are attenuated by premarital teenage fertility, which is high in the region (Lesthaeghe and Jolly, 1995). Still, the timing of entry into marriage affects fertility as most births to women, in the developing countries, occur in marital union (US Bureau of the Census, 1996).

Early marriage not only increases fertility; it can, if accompanied by early childbearing, also endanger the health of the woman and her offspring and thus cause high child and maternal mortality (Acsadi and Johnson-Acsadi, 1990; Gupta and Mahy, 2003). Furthermore, early marriage may also contribute to a reduction in education and employment opportunities for women, hindering them from contributing effectively to the social and economic development of a country (Siu-Man and Boachang, 1995). Education is a significant factor in determining entry into marriage as an increase in educational attainment positively correlates with an increased age at first marriage. This can be explained by the longer period that is spent in schools, but the relationship with timing of entry into marriage is confounded by other factors such as place and type of residence, ethnicity and religion (Maitra, 2004). Birth cohort differentials in age at first marriage provide an indication of trends in marriage rates over time. However, trend analysis is hampered by recall errors among older women, of events that happened many years ago, as well as by under-reporting of current unions in young women (Singh, 1998; Harwood-Lejeune, 2000). Thus, only trends in recent years are worth examining. The data that are available for this study come from the 2000 Malawi Demographic and Health Survey (the 2000 MDHS). The survey used a hierarchically clustered sampling design—in particular at the family and community clustering levels. We examine the socioeconomic and demographic factors influencing the timing of first marriage in Malawi and, in particular, the extent of variation in marriage rates within and across families and communities.

The rest of the paper is organized as follows. Section 2 presents some substantive and methodological considerations in modelling and analysing marriage data with many covariates in a multilevel structure. The study data are introduced in Section 3, in which the covariates and the multilevel structure that is used in the analyses are described. In Section 4, we discuss a three-level discrete survival model which is appropriate for our data, and its Bayesian computation is presented in Section 5. The results from the use of our model on the example data are

contained in Section 6. Finally, a discussion on the model and results is presented in Section 7. The WinBUGS code that was used to analyse the data can be obtained from

<http://www.blackwellpublishing.com/rss>

2. Modelling considerations

As elaborated in the following paragraphs, an appropriate model for this application needs to take account of the discrete measurement of time to marriage. The model should also measure the influence of the covariate factors as well as allow for heterogeneity between families and communities, in addition to investigating possible cross-level association between family and community levels. Finally, rather than constraining the underlying base-line hazard to a certain functional form, we want to allow for flexible modelling of the base-line hazard function.

In demographic studies, proportional hazards regression is used widely to model data on event histories such as age at marriage (Santow and Bracher, 1994). The proportional hazards model is based on the assumption that time is measured on a continuous scale. However, in demography, time is often reported and recorded as grouped continuous. Different true durations are reported to have the same duration, often in discrete time, typically months or years, resulting in a large number of tied time points. Estimation of the model with such heavily tied failure times would generally give inconsistent estimators (Prentice and Gloeckler, 1978). Breaking ties by slight adjustments does not solve the problem as this introduces accuracy bias into the data. In these situations, discrete time duration models are more appropriate. Discrete time survival models treat the indicators of event of interest at each time point as independent binary trials and use a series of binary regression models at each time point (Allison, 1982; Arjas and Haar, 1987; Fahrmeir and Knorr-Held, 1997). The discrete time duration models have been extended to account for unobserved heterogeneity (Lewis and Raftery, 1999; Biggeri *et al.*, 2001), usually including only a random effect at an individual level, except Biggeri *et al.* (2001), who used two independent nested random effects.

Additionally, in a nested sample, individual marital union histories within the same family or community cluster are more likely to have similar cultural beliefs and access to family planning programmes such that their responses will not be independent. Thus, an analysis of such data should control for any possible clustering of the responses (Gupta and Mahy, 2003). An analysis that assumes independence of the outcomes may result in an underestimation of variances (due to the reduced effective sample size) leading to an inflation of the significance of the effects (Goldstein, 1991). Moreover, ignoring clustering in survival data analysis may also produce biased estimates of parameters, especially when the variance of the unobserved clustering effect is large (Guo and Rodriguez, 1992). We also want to investigate cross-level correlation. We want to see whether the effects are amplified when, for example, a family which adheres to traditional customs rigidly resides in a very traditional community. If this is so, then neglecting cross-level correlation may be wrong. Similarly, a poor family may have much more advantage than, say, a mid-level social family as a result of living in a community with low marriage rates. This could happen, if, for example, there was a programme in an urban city to target high risk families with an intervention to decrease fertility levels. An appropriate model that combines both within- and cross-level correlation of the random effects is the multivariate normal model, which has been used in spatial random-effects modelling of multiple-disease outcomes in epidemiology (see for example Leyland *et al.* (2000)).

We model the base-line hazard as piecewise constant but also allow for the possibility that the hazards in two consecutive time points are not completely unrelated to each other. Naturally, we would expect the hazards to change smoothly between time points. Even though a

polynomial function could be used for the base-line hazard (Biggeri *et al.*, 2001), more flexible nonparametric modelling is desired. A dynamic model using a random-walk prior with variance depending on the length of the interval is the natural choice. This type of model does not impose any functional form for dependence of parameters on time, and as such it can be regarded as nonparametric (Fahrmeir and Knorr-Held, 1997). The nonparametric base-line hazard has a greater ability to capture hidden patterns such as peaks and bumps than parametric models have. In using cross-level dependence between family and community frailties, a nonparametric base-line hazard and a discrete time duration model, this paper is a generalization of earlier analyses, and it differs, in particular, from Biggeri *et al.* (2001) in the use of full Bayesian inference using Markov chain Monte Carlo simulation techniques. Bayesian methods for discrete time-to-event models have also been used by Fahrmeir and Knorr-Held (1997) to model employment data from Germany and by Lewis and Raftery (1999) on fertility decline in Iran.

3. The data

The data that were available for analysis in this study were collected in 2000 as part of the MDHS that was conducted by the National Statistical Office. The survey is part of a programme that is conducted in many developing countries to provide cross-sectional information on fertility and child and infant mortality levels, family planning, marriage, partner relations and sexually transmitted diseases, including acquired immune deficiency syndrome. Further details about the programme and how to obtain data sets can be found at <http://www.measuredhs.com>. The 2000 MDHS used six sampling strata: urban and rural domains for each of the three administrative regions of the country: the northern, central and southern regions. The survey employed a two-stage sampling design. In the first stage, a systematic stratified sample of 560 enumeration areas was selected: 449 in rural areas and 111 in urban areas. At the second stage, a sample of households was taken from within each selected enumeration area. 11 districts were deliberately oversampled to ensure that reliable estimates of social-economic and health variables were calculated.

In the households sampled, all women aged 15–49 years were eligible for the interview. 13 220 women were interviewed from 10 787 households distributed over the 560 enumeration areas. For the 2000 MDHS, a household was defined as one or more people related or unrelated living together and sharing food resources, and we take this sampling unit as a family unit. The enumeration areas are census tracts which correspond to villages, and these provide community units. The range and mean numbers of women per family are 1–9 and 1.3; however, about 20% of the families had more than one woman respondent. The interquartile range of the number of women per community ranged from 8 to 29 with mean and medium numbers 24 and 23 respectively. Each woman respondent was asked about all her current and past marital and non-marital unions. Of these women, 10 936 (83%) had been or are currently married with mean and median age at first marriage estimated at 17.5 and 17.9 years respectively. Here age at first marriage is defined as age at formal (traditional) marriage and age when the woman started to live in union with a man (US Bureau of the Census, 1996). Fig. 1 shows the estimated empirical discrete hazard function, which has a steep rise after age 15 years to a peak at ages 18–20 years, followed by a steep decline up to about 25 years, from which the hazard gradually tails off.

We considered four explanatory variables for inclusion in the analyses. The region of residence, whose boundaries largely define ethnic lines, was used to study ethnic differentials, the type of place of residence (whether urban or rural) to capture the effects of urbanization and modernization, the *birth cohort*, categorized as 1951–1955, 1956–1960, 1961–1965, 1966–1970, 1971–1975, 1976–1980 and 1981–1985 to capture the effect of changing time on early marriages

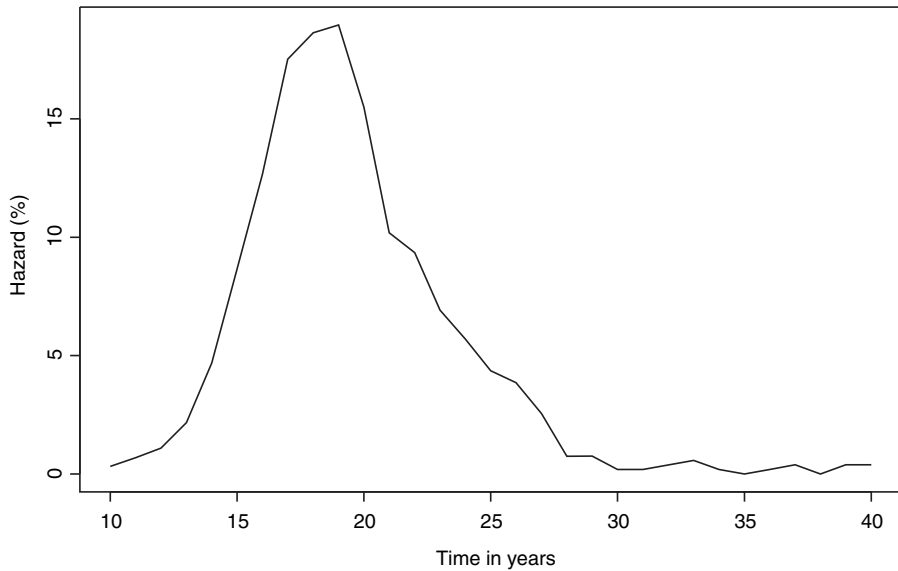


Fig. 1. Empirical hazard function of age at first marriage

and, finally, the level of education of the woman to capture the effects of modernization. A referee rightly pointed out that the inclusion of the woman's education in the hazard regression equation for age at first marriage may result in endogeneity of education. A young woman may leave school after marrying, resulting in reverse causality. We accounted for the possibility of endogeneity of education by using three categories for education of the woman: no education, primary education (up to 8 years of schooling) and secondary education. As in the study of childbearing (Gupta and Mahy, 2003), not many young women in the first 8 years of schooling would be forced to leave school because they married. Formally, a two-stage modelling process could have been used, with the first stage being a regression equation for education, using covariates such as woman's birth cohort, parents' education and region of, and rural-urban, residence. Its predicted value could then be used as an explanatory variable in the hazard model for age at first marriage. This is the approach that was adopted by Maitra (2004) in the study of total fertility, but not here.

Table 1 shows the distribution of the covariates, with the names of the corresponding coefficients. In comparisons with census figures, the urban areas and the northern region were clearly oversampled. However, the results that are presented in this study were not weighted. To investigate marriage rates by birth cohorts, proportions of birth cohorts married before various exact ages were calculated and are shown in Fig. 2. If we consider only births after 1960, then there appears to be a decline in early marriage rates. The proportion of women who were married by age 15 years has declined from 25% among women who were born around 1960 to 10% among women who were born around 1985. It is not possible to quantify reporting errors. However, the MDHS data sets are of reasonable quality such that under-reporting and misreporting errors would not have a major effect on the results (Harwood-Lejeune, 2000).

We examined the existence of variation in marriage rates at various levels of the hierarchy in the data that is still present after controlling for the observed covariates by using family and community random effects. These variations are a result of unobserved or unmeasured explanatory factors. For example, at the family level, these may include the social status and traditional beliefs, which may influence family planning decisions and female independence. At the community

Table 1. Summary of covariates that are used in the model

<i>Variable</i>	<i>Frequency</i>	<i>%</i>	<i>Coefficient</i>
<i>Year of birth</i>			
1981–1985	2914	22.0	0.0
1976–1980	2998	22.7	β_1
1971–1975	2358	17.8	β_2
1966–1970	1574	11.9	β_3
1961–1965	1410	10.7	β_4
1956–1960	1052	8.0	β_5
1951–1955	914	6.9	β_6
<i>Residence</i>			
Urban	2871	21.7	0.0
Rural	10349	78.3	β_7
<i>Region</i>			
Northern	2187	16.5	0.0
Central	4508	34.1	β_8
Southern	6525	49.4	β_9
<i>Education level of woman</i>			
No education	3372	25.5	β_{10}
Primary	8219	62.2	β_{11}
Secondary and higher	1629	12.3	0.0
Total	13220	100.0	

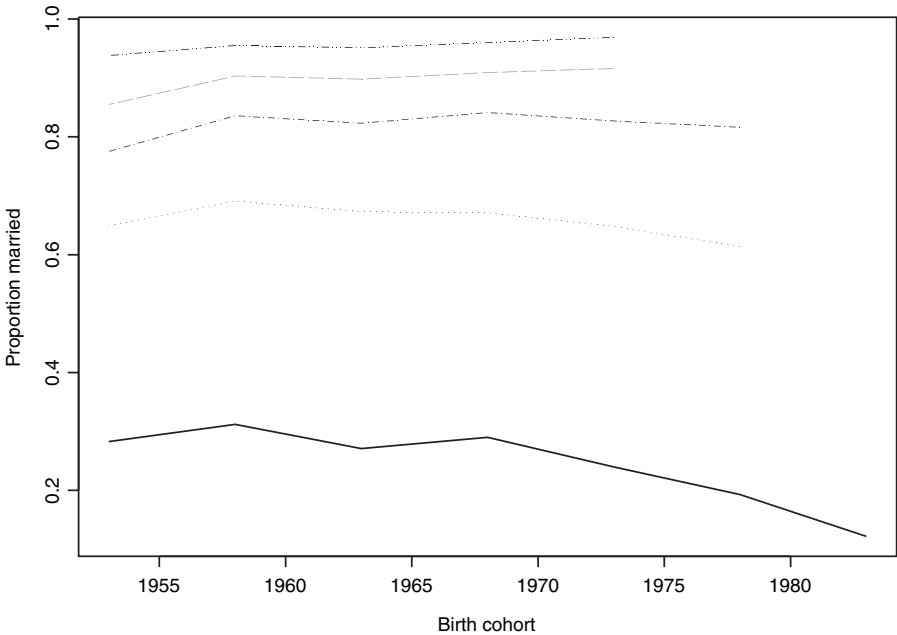


Fig. 2. Proportions (by using life-table estimates) of birth cohorts whose members married by age 15 years (—), 18 years (·····), 20 years (· - · - ·), 22 years (— — —) and 25 years (- · · -)

level, they may include community level education, sanctions against premarital childbirth, cultural approved age at marriage and religious beliefs that would have the effect of increasing or decreasing the timing of marriage (Gupta and Mahy, 2003; Maitra, 2004). Although there are sufficient numbers of women for community level estimation, the data are very sparse for the family effects. Many families have only one woman respondent such that a family unit is indistinguishable from a woman. This should not be a major problem as our Bayesian hierarchical modelling combines data from various families by exploiting the similarity relationship between the parameters. This will provide individual frailty effects in cases where there is only one woman respondent.

4. The model

In discrete time survival analysis, the timescale is partitioned into m intervals $\{I_t = [a_{t-1}, a_t)\}$ with $0 = a_0 < a_1 < \dots < a_m < \infty$. We assume that all censoring happens at the end of the intervals, and the intervals need not necessarily be of equal length. We observe the discrete time $T = t$ with t in $\{1, \dots, m\}$ denoting an observed event in interval I_t . Now consider the k th individual in family j in community i with $x_{ijk(t)}$, a vector of covariates at time t ($i = 1, \dots, N; j = 1, \dots, J_i; k = 1, \dots, K_{ij}$). Conditionally on the unobserved covariates or log-frailties b_{ij} and b_i for family ij and community i respectively, the discrete hazard function for this individual is given by

$$h(t|x_{ijk(t)}, b_{ij}, b_i) = \Pr(T = t | T \geq t; x_{ijk(t)}, b_{ij}, b_i), \quad (1)$$

which is the conditional probability of failure (marriage) in interval $[a_{t-1}, a_t)$ given survival (remaining unmarried) to the beginning of the interval. The corresponding discrete survival function is given by

$$S(t|x_{ijk(t)}, b_{ij}, b_i) = \Pr(T > t | x_{ijk(t)}, b_{ij}, b_i) = \prod_{t=1}^t \{1 - h(t|x_{ijk(t)}, b_{ij}, b_i)\}, \quad (2)$$

which is the probability of surviving beyond a_t . In a more general setting the observed response t is regarded as the expression of an underlying continuous process Y , which is commonly known as the tolerance, with distribution function F . The observed event is time t when the tolerance exceeds a certain threshold value, and a model for the discrete hazard function is usually given by

$$h(t|x_{ijk(t)}, b_{ij}, b_i) = F(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i) \quad (3)$$

where α_t represents a time-varying base-line effect and β is a vector of unknown fixed covariate effects. The extreme value distribution function

$$F(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i) = 1 - \exp\{-\exp(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i)\}$$

and the logistic distribution

$$F(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i) = \frac{\exp(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i)}{1 + \exp(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i)}$$

are often used. The former is derived by assuming Cox's proportional hazards model for the underlying failure times on a continuous scale; hence the fixed effects are the logarithms of the relative risks. The models are similar for short intervals (Thompson, 1977). In this study, we use the logistic model where each component of the parameter vector β is interpreted as the logarithm of the odds ratio of marrying to not marrying for the corresponding fixed effect covariate at each time point, and the random effects b_i and b_{ij} are the community- and

family-specific logarithms of odds of marrying. An α_t represents the logarithm of the base-line odds of marrying at time t .

Now, suppose that woman ijk has observed discrete failure time t_{ijk} . The woman's contribution to the likelihood is given by

$$L_{ijk} = \prod_{t=1}^{t_{ijk}} \frac{\{\exp(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i)\}^{d_{ijk t}}}{1 + \exp(\alpha_t + \beta' x_{ijk(t)} + b_{ij} + b_i)} \quad (4)$$

where $d_{ijk t} = 1$ if she was first married within $[a_{t-1}, a_t)$ years of age and $d_{ijk t} = 0$ otherwise. The contribution is just the likelihood that is obtained by treating the censoring indicators $d_{ijk t}$ as independent Bernoulli trials across discrete time points t (Allison, 1982). The three-level discrete time survival model that is presented here needs advanced estimation methods to estimate the base-line and fixed effects $\alpha_1, \dots, \alpha_m$ and $\beta_1, \dots, \beta_p$ as well as all b_{ij} and b_i and their covariance components. For an analysis of employment data with independent random effects, Biggeri *et al.* (2001) used penalized quasi-likelihood, which is an approximate method. But, as pointed out in Rodríguez and Goldman (2001), these approximate likelihood-type methods result in a substantial underestimation of both fixed and random effects when the random effects are large. In this paper, all the model parameters are estimated by using a Bayesian approach via Markov chain Monte Carlo sampling. We believe that this is a better method to handle the complexity of the model.

5. Bayesian inference

5.1. Setting prior distributions

In Bayesian analysis, the proposed model of the observed data is combined with the prior distribution of all the unknown model parameters. The base-line effects α_t are modelled as piecewise constants, but we also allow for a simple relationship between hazards in two consecutive intervals by using a first-order autocorrelated prior:

$$\alpha_t | \alpha_{t-1} = \alpha_{t-1} + e_t, \quad t = 1, \dots, m, \quad (5)$$

where $\alpha_0 = 0$, $e_t \sim \text{normal}(0, \nu_t^2)$ and e_t is independent of previous α_t s. For variable interval lengths, it seems reasonable to set $\nu_t^2 = \delta_t \nu_0^2$ where $\delta_t = a_t - a_{t-1}$, the length of interval I_t . The joint prior density of the base-line hazards is thus given by

$$\pi(\alpha) \propto (\nu_0^2)^{-m/2} \exp \left\{ -\frac{1}{2\nu_0^2} \sum_{t=1}^m \frac{(\alpha_t - \alpha_{t-1})^2}{\delta_t} \right\}. \quad (6)$$

This is a dynamic model as considered in Gamerman and West (1987). Fahrmeir and Knorr-Held (1997) also allowed the fixed effect vectors to vary over the discretized timescale in the same manner. The fixed effect parameters are modelled as multivariate normal with mean 0 and covariance matrix $d_0 I$ where d_0 was set to 1000, giving a prior which has its mass well spread over the region around 0, but also allows a wide range of other values.

Next, we consider the modelling of the cross-level log-frailties b_{ij} and b_i . These are typically assumed to be independent and identically distributed variables from normal distributions (Biggeri *et al.*, 2001). To capture the cross-level dependence between log-frailties, we assumed a $(J_i + 1)$ -dimensional multivariate normal prior for the random-effects vector $(b_{i1}, b_{i2}, \dots, b_{iJ_i}, b_i)$ with mean 0 and covariance matrix having the first $J_i \times J_i$ elements given by σ_f^2 times the identity matrix and the final row and column given by $(\rho \sigma_f \sigma_c, \dots, \rho \sigma_f \sigma_c, \sigma_c^2)$. By setting the mean to 0, the specific log-frailties are interpreted as deviations from the overall average on the log-odds-

ratio scale. In the actual implementation, we used the marginal bivariate normal density for (b_{ij}, b_i) , which was factored into the product of the marginal density

$$f(b_i|\sigma_c^2) \propto (\sigma_c^2)^{-1/2} \exp\left(-\frac{1}{2\sigma_c^2}b_i^2\right)$$

and the conditional density

$$f(b_{ij}|b_i, \sigma_f^2, \sigma_c^2, \rho) \propto \{\sigma_f^2(1-\rho)\}^{-1/2} \exp\left\{-\frac{1}{\sigma_f^2(1-\rho)}\left(b_{ij}-\frac{\rho\sigma_f}{\sigma_c}b_i\right)^2\right\}.$$

For the variance components, we follow the usual practice of assigning them inverted gamma prior distributions. This is equivalent to specifying a gamma prior distribution for the corresponding precision parameters. Thus, the prior for the precision parameter of the base-line jump ($\tau_0 = 1/\nu_0^2$) has a gamma distribution:

$$f(\tau_0) \propto \tau_0^{\kappa-1} \exp(-\omega\tau_0)$$

where κ and ω are positive parameters which should be specified. Ideally we shall be looking at many intervals with small widths such that large jumps would be very unlikely. So we are looking for a prior which favours small values of ν_0^2 . By setting $\kappa = 0.001$ and $\omega = 0.001$, the prior distribution for τ_0 will favour small values of ν_0^2 but covers other values reasonably well. However, in specifying prior distributions for the variance components $\sigma_c^2 (= 1/\tau_c)$ and $\sigma_f^2 (= 1/\tau_f)$, care should be taken when using very flat or non-informative priors as these can cause problems of convergence and can lead to improper posterior distributions. To guard against these problems, proper priors will be used for the variance components.

We base our prior elicitation on data from 19 sub-Saharan countries (excluding Malawi) on the timing of first births as presented in Singh (1998). The use of evidence from data on first births is appropriate since there is a very strong linear correlation ($\text{corr} = 0.910$) between the time to first birth and the time to first marriage as shown by Acsadi and Johnson-Acsadi (1990) for sub-Saharan and other developing countries, and both are determined by the same factors. 18 of the 19 countries had odds ratio of early childbearing in the range 0.607–3.121 with mean and median values 1.900 and 1.990 respectively, among rural women compared with urban women; the range was 1.084–8.110 with mean and median values 3.77 and 2.900 respectively, among women with less than 7 years of schooling compared with those with 7 or more years of schooling. Clearly place of residence and education are important predictors. After allowing for these factors, we would expect the odds ratio between two families and communities to fall within these ranges, and the random effects to have small variations. The model assumes that 90% of the families have $\log(\text{odds})$ of their women entering marriage in a range with width $2 \times 1.64/\sqrt{\tau_f}$. We also thought it plausible that there was a threefold variation between the odds of two families at the extreme ends of the 90% interval, with a fivefold variation being very unlikely. This implies that 9 is a prior guess for τ_f . This leads to a prior $\tau_f \sim \text{gamma}(9, 1)$, whose implied distribution for σ_f^2 is reasonable, even though it assigns only low probabilities to very low and to very high values. The same prior distribution was also used for τ_c . For the cross-level correlation, there is not enough prior information, so a $\text{uniform}(-1, 1)$ prior is placed on ρ .

We assumed prior independence between all the model parameters. However, the resulting joint posterior is not easily obtainable as it involves complex integrals. We use the Gibbs sampler (Gelfand and Smith, 1990), a Markov chain Monte Carlo method for sampling from a joint distribution. It is widely used in Bayesian statistics to approximate joint posterior distributions involving difficult integrals and is implemented in the WinBUGS computer software

(Spiegelhalter *et al.*, 2000). The algorithm requires that sampling is to be done from all full conditional posterior distributions of the parameters, usually univariate conditional posterior distributions. Each of the univariate full conditional posterior distributions involves only those terms from the joint posterior distribution that are relevant to the parameter under consideration. All the conditionals and sampling methods are automatically determined by the WinBUGS software.

6. Results

The time axis was partitioned into 1-year intervals from 10 to 35, so that the discrete time ranges from 1 to 26 ($=m$). The regression and interval parameters were initially set to 0, frailty variances to 1 and cross-level correlation to 0. Given these initial variance component values, starting values for the frailty terms were generated from their priors. A single chain of 50 000 iterations was run and, using convergence diagnostics based on plots of sampled values and Raftery and Lewis (1992) methods, 10 000 iterations were sufficient for the burn-in period. The remaining 40 000 iterations passed the Heidelberger and Welch stationary and halfwidth tests. Also, the integrated autocorrelation times IACT (Sokal, 1996) were calculated for parameters of interest on the 40 000 iterations. IACT measures the number of correlated samples in the Markov chain Monte Carlo chain having the same efficiency as a single independent sample. The maximum value of IACT was 50, which is reasonable considering the complexity and dimensionality of the parameter space of the model. All post-convergence iterations (10 001–50 000) were retained for posterior summaries.

The estimated mean base-line hazard function is shown in Fig. 3, with 95% credible intervals. The standard deviation of the base-line jump has a posterior mean of 2.038 and standard deviation 0.459. In contrast with a peak hazard at about 18 years in Fig. 1, we now have a peak hazard at 23 years. This reflects the fact that now we are estimating the base-line hazard for

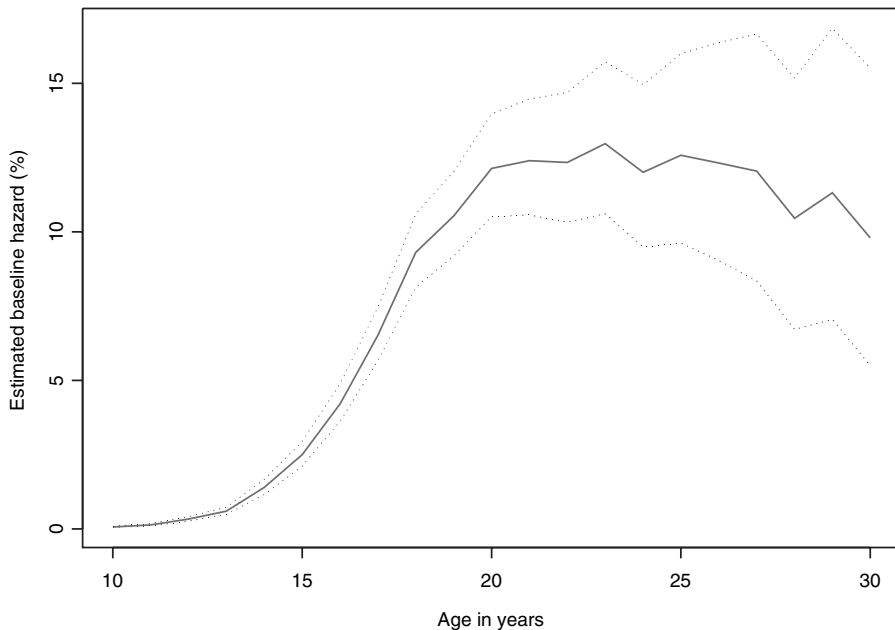


Fig. 3. Estimated mean base-line hazard function with 95% credible interval

Table 2. Comparison of posterior odds ratios of the fixed effect parameters and goodness of fit between random- and fixed effects models

<i>Parameter</i>	<i>Results for the random-effects model</i>			<i>Results for the fixed effects model</i>	
	<i>Mean</i>	<i>Standard deviation</i>	<i>95% credible interval</i>	<i>Mean</i>	<i>Standard deviation</i>
<i>Year of birth</i>					
1981–1985	1.000	—	—	—	—
1976–1980	1.655	0.076	1.512, 1.813	1.588	0.062
1971–1975	1.692	0.081	1.539, 1.851	1.605	0.068
1966–1970	1.708	0.092	1.533, 1.894	1.564	0.069
1961–1965	1.580	0.087	1.412, 1.762	1.437	0.068
1956–1960	1.715	0.106	1.516, 1.936	1.507	0.076
1951–1955	1.311	0.083	1.161, 1.472	1.213	0.062
<i>Residence</i>					
Urban	1.000	—	—	—	—
Rural	1.021	0.042	0.940, 1.098	1.021	0.030
<i>Region</i>					
Northern	1.000	—	—	—	—
Central	0.747	0.036	0.678, 0.818	0.786	0.026
Southern	0.920	0.029	0.837, 1.007	0.923	0.028
<i>Woman's education</i>					
No education	4.065	0.251	3.585, 4.573	3.191	0.148
Primary education	3.781	0.196	3.376, 4.175	3.105	0.129
Secondary and higher	1.000	—	—	—	—
$\bar{D}$	55193			57962	
DIC	57444			57998	
p_D	2251			36	

women in a low risk group, i.e. who were born in 1981–1985, in urban areas, and who obtained secondary education. The hazard function in Fig. 1 is global. Furthermore, we no longer see a decline in the base-line hazard after the peak. This is because we have controlled for unobserved heterogeneity and thus precluded the usual selection effect whereby the ‘early marrying types’ are gradually eliminated from the risk set so that the population that is still at risk of marrying at older ages is increasingly composed of ‘non-marrying types’.

The posterior means, standard deviation and 95% credible intervals for the odds ratios of the fixed effects from fitting the random-effects model (3) are given in the second, third and fourth columns respectively of Table 2. Overall, the odds of marrying show a downward trend: compared with women who were born after 1981, the odds of marrying among women who were born in the years 1966–1970, 1971–1975 and 1976–1980 are all about two times higher. However, only 37% of the women who were born after 1981 were ever married at the time of the survey. Education was also found to have a significant effect on raising the age at first marriage. The odds of marrying among women with less than 8 years of education (primary) are about four times higher than the odds among women with more than 8 years of education (secondary and higher). Region was also found to be an important determinant of rates of marriage. Place of residence, though not significant, shows that marriage rates among rural women are increased by 2%. Posterior correlations between the fixed effects parameters were also

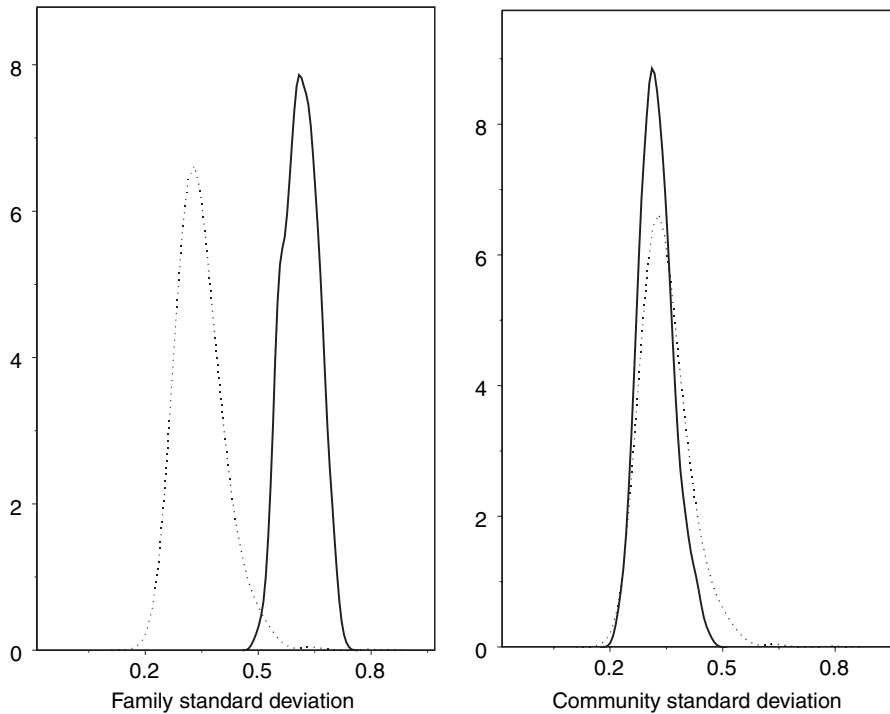


Fig. 4. Prior (·····) and posterior (—) densities for the random-effect standard deviations

examined, and the strongly related were no education and primary education ($\text{corr} = 0.750$), central and southern regions ($\text{corr} = 0.633$) and between birth cohorts estimates ($0.403 \leq \text{corr} \leq 0.635$). An interaction term between urban–rural and education effects was also added to the model to examine the effect of differential education levels of women in urban and rural areas. The two interaction fixed effect terms had 95% confidence intervals $(-0.362, 0.037)$ and $(-0.233, 0.093)$, showing that both were not significant. Thus, differential levels of education in the urban and rural areas do not affect the substantive conclusions concerning education effects.

Fig. 4 shows prior and posterior densities of the standard deviations of the random effects, and it indicates that the family variation is more affected by the data than that of the community. The posterior median and 90% credible intervals for the family and community effect standard deviations are 0.613 (0.540, 0.687) and 0.320 (0.255, 0.407) respectively. The cross-level correlation has posterior median -0.252 and a 90% credible interval $(-0.384, -0.122)$. We also calculated the posterior range of variation in the odds of marriage across families or communities that is not explained by the covariates in the model. For two families, there was about a sevenfold variation between the upper and lower ends of a 90% interval, and for two communities the variation was threefold, which was similar to the prior variation. Lewis and Raftery (1999) derived an inverted gamma(0.470, 0.023) prior distribution for the variance of the women-specific random effect in the context of an Iran fertility analysis. Using this prior for the variance components, the posterior summaries of the fixed effects parameters were essentially the same as those which were obtained with our elicited prior. There was a slight difference in the random-effect variance and correlation parameters, which now have posterior medians (and 90% credible intervals in parentheses) 0.675 (0.579, 0.733), 0.271 (0.092, 0.312) and -0.15 ($-0.343, -0.046$) respectively for standard deviations of the family and community random

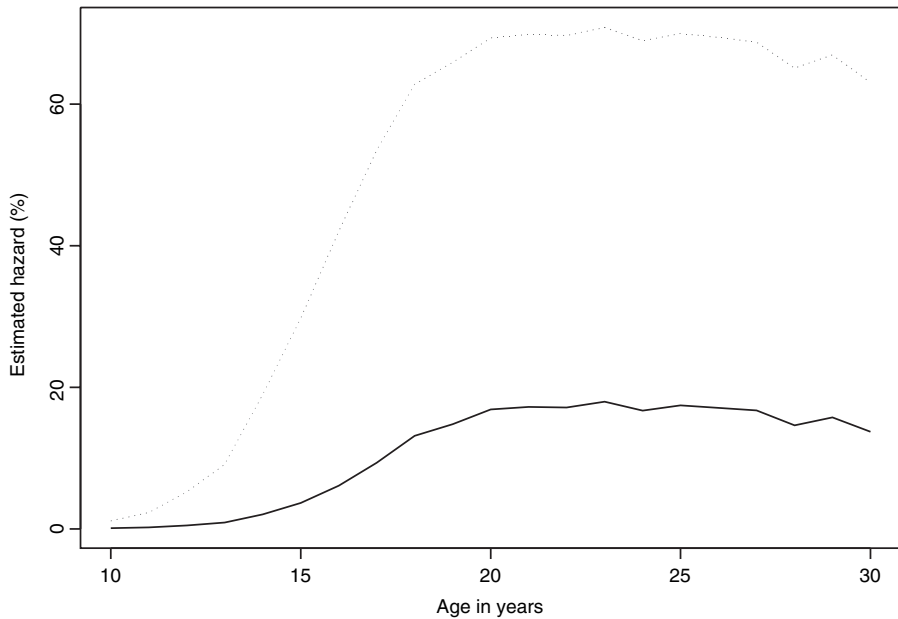


Fig. 5. Estimated mean hazard function for a woman in the lower (—) and upper (·····) 5% of the family risks

effects and the cross-correlation. This shows that the estimates of random-effect parameters are all significant, with more variations due to the family effect than to the community effect. Thus, variations in entry to marriage depend more on differences at the family level than at the community level. The cross-level correlation indicates that a family with a high marriage rate may as well reside in a community with a generally lower rate of marriage. The significance of the family random effect can be better explained by plotting the hazards for two women, with identical covariates but values of random effects at the opposite ends of the range. Fig. 5 shows the hazards for two women who were born between 1966 and 1970, residing in the rural area of the central region, and with primary education, but their family random effects are at the opposite ends of a 90% confidence interval.

In a model where the random-effect parameters are significant, inference of the risk factors is not as simple as in the logistic model without random effects where the exponents of fixed effect parameters are the odds ratios of marrying. Now the exponents must be adjusted by a magnitude depending on the family and community effects. However, we can consider families and communities with specified random-effect values. Fig. 6 shows estimated hazard functions for women in various levels of education with other covariates and random effects set to 0. Women with secondary education have smaller hazards in all age intervals except in very low ages.

We also implemented the fixed effects model without the random-family and community effects. For comparison, the posterior means and standard deviations of the fixed effects are presented in the last two columns of Table 2. The fixed-effects-only model produces estimates of both the odds ratios and the standard deviation that are smaller in magnitude. This is expected because the fixed-effects-only model ignores the correlation between observations in the same family and in the same community, which underestimates the standard errors. Thus, using the fixed-effects-only model, the results would yield credible intervals that are narrower than they should be and may lead to false conclusions. To compare model complexities and goodness of fit formally, the recently proposed deviance information criterion (DIC) of Spiegelhalter

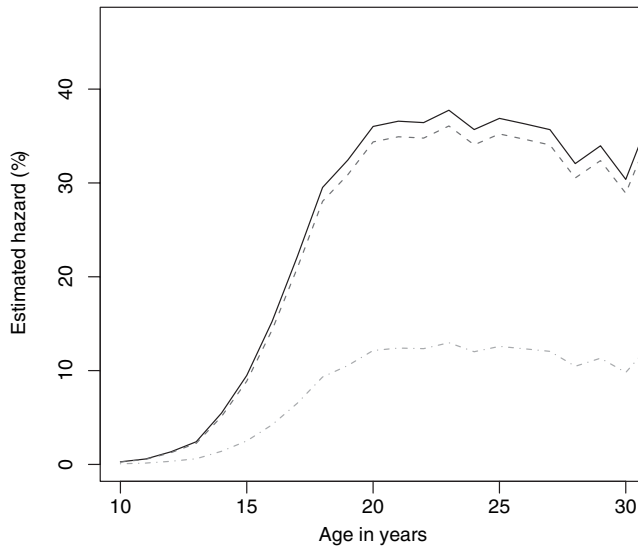


Fig. 6. Estimated mean hazard function by woman's education with the other covariates and random effects set to 0; —, no education; ----, primary education; · - · - ·, secondary and higher education

et al. (2002) was used. The DIC is particularly useful in situations involving complex hierarchical models in which it is difficult to compute the actual number of parameters being used. It is a generalization of the Akaike information criterion (AIC) and works on similar principles. The DIC consists of a measure of *fit* of a model, the posterior expectation of the deviance $\bar{D}$, and a term penalizing the complexity of a model, the *effective number of parameters* p_D . Overall, by including the random effects, the DIC is considerably reduced from 57998 to 57444 even though the random-effects model is far more complex. More importantly, we obtain a substantial improvement in the pure measure of fit, $\bar{D}$, down from 57962 to 55193. We also implemented the constant base-line hazard model and compared its fit to the data. The parameter estimates were in general agreement with those from the other two models. We obtained a DIC value of 57456 as opposed to 57444 for the autoregressive AR(1) model, which means that the AR(1) model gives an overall better fit by reducing the effective number of parameters from 2267 to 2251.

7. Conclusion

This paper introduces a Bayesian hierarchical model for the analysis of discrete time survival data. The data concern age at first marriage among women in Malawi. This is an important application as age at first marriage defines the exposure-to-risk period of childbearing, and hence determines the fertility of a society. Besides modelling the influence of explanatory variables, the model allows for unobserved heterogeneity at two levels of hierarchy as well as cross-level association. Rather than making rigid parametric assumptions, the model also includes a flexible nonparametric prior for the discrete base-line hazard. The model is fitted by using the Gibbs sampler, a simulation-based Bayesian computational algorithm. The implementation is simple and effective, and provides a sample from the joint posterior distributions of all the model parameters. In contrast with maximum-likelihood-based techniques that provide only point estimates, we can employ the sample from the joint posterior to explore the behaviour of parameters and any function of the parameters of interest.

Women with higher levels of educational attainment are far more likely to enter marriage at later ages than those without any or with little education. There were also substantial birth cohort differences, with members of younger cohorts delaying marriage as opposed to older women. These findings are consistent with results obtained by Singh (1998) and Gupta and Mahy (2003). The US Bureau of the Census (1996) conducted a comparative study of marriage rates in sub-Saharan Africa that showed that proportions of teenage marriages are declining, which is confirmed by our findings since women born in 1981–1985 were still teenagers by the year 2000 when this MDHS was conducted. The observed differentials in education operate through other elements of socioeconomic development that are associated with education. Educated women are an urbanized and modernized subgroup with better socioeconomic status conducive towards delayed marriages (Maitra, 2004). Moreover, they are more likely to use family planning within marriage to limit and space out their children to develop careers (US Bureau of the Census, 1996). Although improvements have been achieved, the levels of women's education in the rural areas of the sub-Sahara region remain low (Kirk and Pillet, 1998). Thus, the status of education among rural women in the region indicates that the environment is not conducive to any significant increase in age at first marriage. The Government of Malawi has acknowledged the importance of improving women's education to curb the negative social and economic consequences of early marriage. Programmes that strengthen education, employment and family planning have been embarked on. These programmes need strong and sustained commitment by the Government, a task that has been made even more difficult by the adverse economic conditions and resources (Caldwell and Caldwell, 2003).

It has also been observed that, besides strong education and birth cohort effects on the differences in age at first marriage in Malawi, the unmeasured family level factors that are captured by the family level random effect actually account for by far the largest proportion of the variation in age at first marriage (this is clearly seen by comparing the hazards in Figs 5 and 6, where the hazards between families vary between about 10% and 60%, whereas the hazards between different levels of education vary between about 10% and 45% for the women who are over 20 years old). The unobserved family and community heterogeneity are important as family and community level covariates have not been included in the model. The unobserved family level factors which could affect age at marriage include ethnicity, religion and parental education (Maitra, 2004). The unobserved community heterogeneity takes account of community level education, urban–rural residence, family planning and reproductive health facilities, whose access depends on the community's physical infrastructure and cultural norms (Gupta and Mahy, 2003). In communities where premarital childbearing is not socially and culturally accepted, a rising age at first marriage may play a crucial role in the transition of fertility from high to low levels (Maitra, 2004). The strength of both variations implies that targeting both families and communities in a programme to reduce fertility levels would bear fruit. It has also been shown that there is negative association between the family effects and community effects. In terms of policies, this implies that targeting communities with an intervention to increase age at first marriage may not necessarily benefit individual families in lowering levels of fertility. A campaign that is geared towards individual families would achieve the desired goals.

In Malawi, the levels of fertility are high and there is an encouragement for early reproduction and marriage especially in rural areas (Kavinya, 2001). Early marriage and motherhood are the traditional ways in which a woman affirms her value and identity within a family and community. Under these conditions, even exposure to formal education may unexpectedly lead to high fertility, as education itself may undermine traditional practices that are aimed at controlling fertility (Gupta and Mahy, 2003). Thus, policy makers have the task and responsibility of changing the social climate by initiating education programmes that point out the advantages

of delayed marriage. This might involve, for example, building a public consensus about an acceptable marriageable age. Acsadi and Johnson-Acsadi (1990) argued for marriage laws, with proper sanctions and enforcement, which in practice might be very difficult to implement in many rural areas of the region. All efforts that are geared at increasing ages at marriage would only bear minimal positive results unless the levels of women's education and opportunities of employment are greatly improved (Kirk and Pillet, 1998).

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